

**Vishaali B 251001304-2025-IT** ▾**V2****Started on** Thursday, 17 September 2026, 10:51 PM**State** Finished**Completed on** Thursday, 17 September 2026, 10:58 PM**Time taken** 6 mins 34 secs**Marks** 1.00/1.00**Grade** **10.00** out of 10.00 (**100%**)

Question 1 | Correct Mark 1.00 out of 1.00

Convert the following algorithm into a program and find its time complexity using counter method.

```
void reverse(int n)
{
    int rev = 0, remainder;
    while (n != 0)
    {
        remainder = n % 10;
        rev = rev * 10 + remainder;
        n /= 10;
    }
    print(rev);
}
```

Note: No need of counter increment for declarations and scanf() and count variable printf() statements.

Input:

A positive Integer n

Output:

Print the value of the counter variable

Answer:

```
1  #include <stdio.h>
2
3  int main() {
4      int n, rev = 0, remainder, count = 0;
5
6      if (scanf("%d", &n) != 1) return 0;
7
8      count++;
9
10     while (n != 0) {
11         count++;
12
13         count++;
14         remainder = n % 10;
15
16         count++;
17         rev = rev * 10 + remainder;
18
19         count++;
20         n /= 10;
21     }
22
23     count++;
24     count++;
25     printf("%d", count);
26     return 0;
27 }
```

	Input	Expected	Got	
✓	12	11	11	✓
✓	1234	19	19	✓

Passed all tests! ✓

Correct

Marks for this submission: 1.00/1.00.

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